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Roll No.

221044

**4th Sem / ECE, ECE(For Speech
and Hearing Impaired)
Subject : Power Electronics**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 A single Phase full wave fully controlled bridge converter uses (CO3)

- | | |
|----------|------------|
| a) 4SCRs | b) 6 SCR's |
| c) 2SCRs | d) 3SCRs |

Q.2 An inverter converts (CO5)

- a) dc into variable dc
- b) ac into dc
- c) dc into ac
- d) ac into ac of different frequency

Q.3 According to the basics of phase control (CO3)

- a) control is achieved by adjusting the number of on off cycles

- b) control is achieved by adjusting the firing angle of the devices
- c) the load is on for some cycles and off for some cycles
- d) control cannot be achieved

Q.4 The single phase mid-point type cyclo converter uses _____ number of SCRs. (CO3)

- a) 4
- b) 8
- c) 6
- d) none of the mentioned

Q.5 If gate current is increased, the anode-cathode voltage at which SCR conducts is _____ (CO1)

- a) Increased
- b) Decreased
- c) Maximum
- d) Least

Q.6 Chopper converts _____ (CO5)

- a) AC to DC
- b) DC to AC
- c) DC to DC
- d) AC to sawtooth

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 SCR is uni-directional in nature. (True/False) (CO1)

- Q.8 Cycloconverters are used in power supply of air crafts. (True/False) (CO5)
- Q.9 A series inverter uses _____ circuit for commutation of SCR. (CO5)
- Q.10 Define duty cycle of a step down chopper. (CO 5)
- Q.11 UJT is a member of thyristor family. (True/False) (CO2)
- Q.12 How many modes of operation are there for a TRIAC? (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is a closed loop electric drive? (CO5)
- Q.14 Write any two disadvantages of AC drives. (CO5)
- Q.15 Write any four applications of cycloconverters. (CO5)
- Q.16 Why OFF line Ups is called as short break UPS? (CO4)
- Q.17 Explain the basic components of a solar power plant. (CO4)
- Q.18 Draw and explain the VI characteristics of SCR.

- Q.19 Draw layer diagram and two transistor equivalent of an SCR. (CO1)
- Q.20 Explain the light triggering of LASCR. (CO1)
- Q.21 What role heat sink plays in Power Electronics? (CO2)
- Q.22 Explain the working of single phase wave controlled rectifier with resistive load. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the working of online UPS with the help of block diagram. Why it is called online UPS? (8,2)(CO4)
- Q.24 Explain in detail the working of UJT based relaxation oscillator circuit. (CO2)
- Q.25 Describe the construction and working principle of TRIAC with its V-I characteristics. (CO2)